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6 A metal sphere is suspended vertically by an insulating thread. The sphere is charged. The charge causes an electric field around the sphere.

(a) Complete the definition of an electric field.

An electric field is a region where .....

experiences ..... .

[2]

(b) A negatively charged metal rod is brought close to the sphere. Figure 6.1 shows the charged rod and charged sphere.

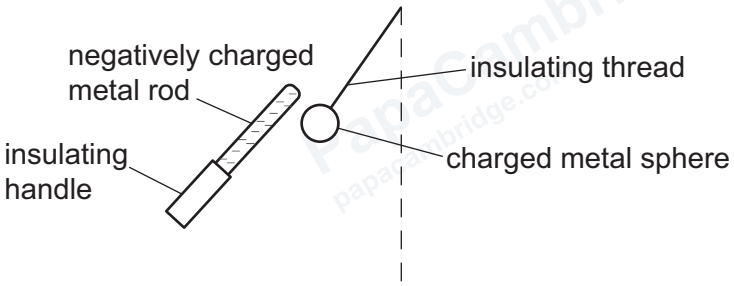

Figure 6.1

(i) State and explain whether the charge on the metal sphere is positive or negative.

statement .....

explanation .....

.....

[1]

(ii) The metal sphere touches the rod.

State what happens to the sphere. Explain this in terms of the movement of particles.

statement .....

explanation .....

.....

.....

[3]

[Total: 6]